

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Two-variable data: Models and scatterplots	Medium

Question

In which of the following tables is the relationship between the values of x and their corresponding y-values nonlinear?

Answer

A.

x	1	2	3	4
y	8	11	14	17

B.

x	1	2	3	4
y	4	8	12	16

C.

x	1	2	3	4
y	8	13	18	23

D.

x	1	2	3	4
y	6	12	24	48

Correct Answer: D

Rationale

Choice D is correct. The relationship between the values of x and their corresponding y-values is nonlinear if the rate of change between these pairs of values isn't constant. The table for choice D gives four pairs of values: (1,6), (2,12), (3,24), and (4,48). Finding the rate of change, or slope, between (1,6) and (2,12) by using the slope formula, $\frac{y_2 - y_1}{x_2 - x_1}$, yields $\frac{12 - 6}{2 - 1}$, or 6. Finding the rate of change between (2,12) and (3,24) yields $\frac{24 - 12}{3 - 2}$, or 12. Finding the rate of change between (3,24) and (4,48) yields $\frac{48 - 24}{4 - 3}$, or 24. Since the rate of change isn't constant for these pairs of values, this table shows a nonlinear relationship.

Choices A, B, and C are incorrect. The rate of change between the values of x and their corresponding y-values in each of these tables is constant, being 3, 4, and 5, respectively. Therefore, each of these tables shows a linear relationship.